A DISTANT-THEOREM AUDIT FOR POINTWISE SQUARE-ROOT CANCELLATION IN MÖBIUS SUMS

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Abstract. We test whether square-root theorems from probability, Boolean analysis, automatic sequences, frame theory, dynamical systems, empirical-process theory, discrepancy, random multiplicative chaos, and function fields can be transferred to the deterministic summatory Möbius function. The audit uses one criterion throughout: an exact representation of the constant vector, a uniform square-root estimate for every representing kernel, and an inverse of total-variation or ℓ^1 norm $N^{o(1)}$. We prove elementary diagnostic results for positive averaging, random centering, point evaluation in ambient L^p spaces, and direct ℓ^1 reconstruction. We also compute the mean predicted by an unrestricted independent p-adic surrogate; it is of order $N/(\log N)^2$, not of square-root order. The reported finite computation on 455,052,511 supplied Möbius values exhibits a large mismatch with that surrogate at the final prefix. None of the theorems in the audited collection currently supplies the complete transfer criterion under the exclusions imposed here. Automatic recursions, function-field specialization, signed stable sampling, and deterministic transfer operators are therefore not retained as completed routes in this audit. A sign-reversing involution with a square-root exceptional set would imply the target estimate directly, whereas the one-prime matching graph has an obstruction of order $N/\log N$. Moreover, the exact orthogonal decomposition of cyclic-window energy contains $|\mathcal{M}(N)|^2$ as its constant-mode component. Thus an uncentered square-root energy estimate already contains the desired conclusion, while a centered estimate removes precisely that component.

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Remark 0.1 (Status of the document). The statements in theorem-like environments are proved in the text. Claims about the literature are cited. The numerical values in Section 12 are reported from the supplied computational record; the underlying binary data and generating script were not part of the present textual re-audit, so those values are not independently recomputed here.

1. Objective and scope

Let

$$\mathcal{M}(N) = \sum_{n \le N} \mu(n). \tag{1.1}$$

The target scale is

$$|\mathcal{M}(N)| \ll_\varepsilon N^{1/2+\varepsilon} \qquad (\varepsilon > 0). \tag{1.2}$$

This report deliberately avoids using classical character methods, explicit formulae, zero-density estimates, classical large-sieve arguments, and nilsequence technology as candidate engines. Those subjects are already adjacent to the classical problem. The purpose is narrower and more severe: locate a theorem proved for a genuinely different object and determine whether its hypotheses can be applied to μ through an exact and stable representation.

The central conclusion is negative but substantive. Every route in the audited collection fails at a named mathematical interface. In most cases the failure is not that the available exponent is slightly too weak. The theorem controls a fluctuation around an unknown mean, a random input rather than a fixed one, a coloring that may be chosen rather than the prescribed Möbius coloring, or an inverse whose norm destroys the square-root gain.

An earlier version retained frame reconstruction and deterministic transfer operators as conditional templates. That classification was too permissive. Once character estimates, large-sieve methods, zero-density arguments, and their close analogues are excluded from the search, neither template supplies an independent mechanism for its required uniform Möbius correlation bound. Section 14 closes both templates and the proposed sign-reversing-involution reformulation within the declared scope.

2. The admissibility criterion

For each N, let $(\Xi_N, \mathcal{A}_N)$ be a measurable space and let $K_N(n,\xi)$ be a measurable kernel for $1 \le n \le N$. A finite complex measure ν_N on Ξ_N gives an exact reconstruction when

$$1 = \int_{\Xi_N} K_N(n,\xi)\, d\nu_N(\xi) \qquad (1 \le n \le N). \tag{2.1}$$

Theorem 2.1 (Exact stable-transfer lemma)**.** *Suppose that* (2.1) *holds and that, for every* $\varepsilon > 0$*,*

$$\sup_{\xi \in \Xi_N} \left| \sum_{n \le N} \mu(n) K_N(n,\xi) \right| \le C_\varepsilon N^{1/2+\varepsilon} \tag{2.2}$$

and

$$\|\nu_N\|_{\mathrm{TV}} \le N^{o(1)}. \tag{2.3}$$

Then (1.2) *holds.*

Proof. Finite summation and integration may be interchanged. Hence

$$\begin{aligned}\mathcal{M}(N) &= \sum_{n\le N}\mu(n)\int_{\Xi_N} K_N(n,\xi)\,d\nu_N(\xi)\\ &= \int_{\Xi_N}\sum_{n\le N}\mu(n)K_N(n,\xi)\,d\nu_N(\xi).\end{aligned} \tag{2.4}$$

Fix $\varepsilon>0$. Apply (2.2) with $\varepsilon/2$. By (2.3), for all sufficiently large N, $\|\nu_N\|_{\mathrm{TV}}\le N^{\varepsilon/2}$. Therefore

$$|\mathcal{M}(N)|\le C_{\varepsilon/2}N^{1/2+\varepsilon}, \tag{2.5}$$

which is (1.2). □

Approximate reconstruction is much less forgiving in the uniform norm.

Proposition 2.2 (Cost of reconstruction error)**.** *Let*

$$e_N(n)=1-\int_{\Xi_N}K_N(n,\xi)\,d\nu_N(\xi). \tag{2.6}$$

Under (2.2)*,*

$$|\mathcal{M}(N)|\le C_\varepsilon N^{1/2+\varepsilon}\|\nu_N\|_{\mathrm{TV}}+\sqrt{N}\,\|e_N\|_2. \tag{2.7}$$

In particular, the elementary estimate from a uniform error $\|e_N\|_\infty\le\delta_N$ contributes at most $N\delta_N$.

Proof. Insert (2.6) into (1.1). The integral is bounded as in Theorem 2.1. Cauchy–Schwarz and $\sum_{n\le N}|\mu(n)|^2\le N$ give the second term. The uniform estimate follows from $\|e_N\|_2\le\sqrt{N}\delta_N$. □

The theorem is elementary, but it prevents a common logical substitution. An average square-root theorem is useful only after the constant vector has been reconstructed with a norm that does not consume the gain.

3. Four universal obstructions

3.1. **Positive averages cannot hide the constant mode.**

Proposition 3.1 (Extreme-point obstruction)**.** *Let ν be a probability measure and suppose $|K(n,\xi)|\le 1$ for all n,ξ. If*

$$1=\int K(n,\xi)\,d\nu(\xi) \tag{3.1}$$

for a fixed n, then $K(n,\xi)=1$ for ν-almost every ξ. Consequently, an exact positive average of unit-disc oscillatory kernels is trivial on every reconstructed coordinate.

Proof. The real part satisfies $\Re K(n,\xi)\le 1$, while its integral is one. Thus $\Re K(n,\xi)=1$ almost surely. The unit-disc condition then forces $K(n,\xi)=1$ almost surely. □

Nontrivial reconstruction therefore requires signed or complex weights, or kernels whose amplitudes exceed one. Each alternative makes the inverse norm an essential part of the proof.

3.2. **Randomization controls fluctuations, not the fixed endpoint.**

Proposition 3.2 (Permutation-centering obstruction)**.** *Fix real numbers $a_1, \dots, a_N$, let Π be a uniformly random permutation, and put*

$$S_m(\Pi) = \sum_{j=1}^{m} a_{\Pi(j)}, \qquad \bar{a} = N^{-1} \sum_{j=1}^{N} a_j. \tag{3.2}$$

Then

$$\mathbb{E}S_m(\Pi) = m\bar{a}, \qquad S_N(\Pi) - \mathbb{E}S_N(\Pi) = 0 \tag{3.3}$$

identically. Therefore no concentration, central-limit, or strong-approximation theorem for the centered random process can bound $\sum_{j=1}^{N} a_j$.

Proof. Every a_j appears at each position with probability $1/N$, which gives the first equality. Every permutation uses every entry exactly once, so $S_N(\Pi) = \sum_j a_j = N\bar{a}$, proving the second equality. □

For $a_j = \mu(j)$, the unknown mean is exactly $\mathcal{M}(N)/N$. Random sampling without replacement can estimate that mean if sampled values are treated as data, but it cannot prove a deterministic upper bound for the population total from the same population.

3.3. **A random-phase moment bound has no bounded deterministic evaluation.**

Proposition 3.3 (Ambient point-evaluation obstruction)**.** *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $\omega_0 \in \Omega$. Suppose that there are measurable sets $A_k \ni \omega_0$ with $\mathbb{P}(A_k) \to 0$. Point evaluation is not well defined on $L^p(\Omega)$ equivalence classes. Moreover, on any linear space of chosen measurable representatives that contains the normalized indicators below, every linear functional agreeing with evaluation at ω_0 is unbounded with respect to the ambient L^p norm, for $1 \le p < \infty$.*

Proof. Take

$$f_k = \mathbb{P}(A_k)^{-1/p} \mathbf{1}_{A_k}. \tag{3.4}$$

Then $\|f_k\|_p = 1$, whereas $|f_k(\omega_0)| = \mathbb{P}(A_k)^{-1/p} \to \infty$. Thus no such evaluation functional is bounded in the ambient L^p norm. The preliminary statement follows because changing a representative on a null set can change its value at ω_0. □

Remark 3.4. Proposition 3.3 is an ambient-space statement. It does not exclude a bounded evaluation theorem on a smaller regularity class, such as a finite-dimensional polynomial space equipped with a stronger norm. A transfer argument based on random phases would need precisely such a restricted theorem, with constants uniform in the arithmetic parameters.

Corollary 3.5 (Exact loss on a finite Rademacher cube)**.** *Let $\Omega_m = \{-1, 1\}^m$ carry the uniform probability measure, fix $\omega_0 \in \Omega_m$, and let $E_{\omega_0} f = f(\omega_0)$. For every $1 \le p < \infty$,*

$$\|E_{\omega_0}\|_{L^p(\Omega_m) \to \mathbb{C}} = 2^{m/p}. \tag{3.5}$$

In particular, passing from an L^2 average over m Rademacher prime phases to one prescribed configuration has optimal worst-case loss $2^{m/2}$.

Proof. For every $f : \Omega_m \to \mathbb{C}$,

$$\|f\|_p^p = 2^{-m} \sum_{\omega \in \Omega_m} |f(\omega)|^p \ge 2^{-m} |f(\omega_0)|^p. \tag{3.6}$$

Thus $|E_{\omega_0} f| \le 2^{m/p} \|f\|_p$. Equality holds for every nonzero function supported on the singleton $\{\omega_0\}$, proving (3.5). □

For the unrestricted space of all functions on the prime-sign cube with one coordinate for every $p \le N$, one has $m = \pi(N)$, so the ambient worst-case norm is $2^{\pi(N)/2}$. This is not $N^{o(1)}$, since the prime number theorem gives $\pi(N) \sim N/\log N$ [17]. The statement is not a claim that this ambient norm is sharp on every restricted arithmetic polynomial subspace.

3.4. **Hilbert-space stability is not the required stability.**

Proposition 3.6 (The ℓ^1 reconstruction cost)**.** *Let $u_1, \dots, u_m \in \mathbb{C}^N$, let $\mathbf{1}_N = \sum_{j=1}^m c_j u_j$, and suppose $\|u_j\|_2 \le L$. Then*

$$\|c\|_1 \ge \frac{\sqrt{N}}{L}. \tag{3.7}$$

Moreover, if $|\langle a, u_j\rangle| \le B$ for all j, the direct reconstruction argument yields only

$$|\langle a, \mathbf{1}_N\rangle| \le B\|c\|_1. \tag{3.8}$$

An ℓ^2 inverse bound does not replace the ℓ^1 quantity in (3.8)*.*

Proof. The triangle inequality gives $\sqrt{N} = \|\mathbf{1}_N\|_2 \le \sum_j |c_j| \|u_j\|_2 \le L\|c\|_1$. The second assertion follows by pairing the reconstruction with a and using the triangle inequality. □

Kashin representations and restricted-invertibility theorems are powerful ℓ^2 stability results. They do not, by themselves, supply the ℓ^1 coefficient bound used in the direct triangle-inequality argument (3.8). An ℓ^2 reconstruction could still be useful if accompanied by a matching ℓ^2 estimate for the correlation vector; no such estimate is obtained from Proposition 3.6.

4. Martingales and exchangeable pairs

Gordin's method and its successors transfer central-limit and invariance principles from martingale differences to stationary processes [6, 10, 23]. Chatterjee's exchangeable-pair method derives concentration from an antisymmetric coupling and a regression structure [4]. These are genuine square-root theories, but their centering is structural rather than cosmetic.

Proposition 4.1 (Constant inputs have zero martingale differences)**.** *Regard each deterministic number a_n as a constant random variable on a probability space, adapted to any filtration. Then*

$$a_n - \mathbb{E}(a_n \mid \mathcal{F}_{n-1}) = 0. \tag{4.1}$$

Hence the canonical martingale-difference part of this constant-random-variable encoding is zero.

Proof. A constant random variable is measurable with respect to every sigma-field and its conditional expectation equals the same constant. □

Randomly permuting the sequence creates nonzero intermediate fluctuations, but Proposition 3.2 shows that the endpoint fluctuation is still zero. Multiplying the fixed values by independent mean-zero signs creates a martingale-difference model under the natural filtration, but changes μ into a different object. An exchangeable prime-factor move would be relevant only after proving a regression identity whose remainder is small uniformly at the fixed Möbius configuration. No such identity is supplied by the exchangeable-pair theorem; it would be the new arithmetic input.

The Maxwell–Woodroofe condition used by Peligrad and Utev contains norms of conditional expectations of partial sums. For a proposed deterministic encoding of Möbius, verifying those conditional expectations at the required scale is not automatic and may contain the original cancellation problem. The route is therefore conditional, not an application of an existing theorem.

5. Boolean hypercontractivity and prime divisibility

For a squarefree integer,

$$\mu(n) = (-1)^{\sum_{p\le N} \mathbf{1}_{p\mid n}}. \tag{5.1}$$

This resembles a Walsh character on a prime-divisibility cube. Bonami–Beckner hypercontractivity controls polynomial norms on product spaces [2, 1]. The resemblance fails in two independent ways: the observable is a high-degree parity character, and the divisibility vectors of integers in an interval are not independent at the resolution needed for that parity.

The unrestricted independent p-adic surrogate makes the discrepancy quantitative. It is not the factorization law of an integer chosen uniformly from $[1, N]$; it deliberately discards the global size constraint. Let $(V_p)_p$ be independent with

$$\mathbb{P}(V_p = j) = (1 - p^{-1})p^{-j}, \qquad j \ge 0, \tag{5.2}$$

and extend $\mu(p^{V_p})$ multiplicatively.

Theorem 5.1 (Mean of the independent p-adic surrogate)**.** *For the model* (5.2),

$$\mathbb{E}\mu(p^{V_p}) = (1 - p^{-1})^2 \tag{5.3}$$

and therefore

$$\mathbb{E} \prod_{p\le N} \mu(p^{V_p}) = \prod_{p\le N} (1 - p^{-1})^2 \sim \frac{e^{-2\gamma}}{(\log N)^2}. \tag{5.4}$$

Consequently, an independent-prime population of N observations has mean of order $N/(\log N)^2$, which exceeds $N^{1/2+\varepsilon}$ for every fixed $\varepsilon < 1/2$ and all sufficiently large N.

Proof. Only $V_p = 0$ and $V_p = 1$ contribute. Hence

$$\begin{aligned} \mathbb{E}\mu(p^{V_p}) &= \mathbb{P}(V_p = 0) - \mathbb{P}(V_p = 1) \\ &= (1 - p^{-1}) - (p^{-1} - p^{-2}) = (1 - p^{-1})^2. \end{aligned} \tag{5.5}$$

Independence gives the product. Mertens' product theorem yields the asymptotic in (5.4); see [17]. Finally, $N^{1/2-\varepsilon}/(\log N)^2 \to \infty$. □

Thus the unrestricted product surrogate does not reproduce the mean behavior of the finite integer prefixes. This diagnoses the loss caused by discarding the global size and divisibility constraints. It does not prove that every Boolean-analytic formulation must fail; a relevant theorem would have to retain those arithmetic constraints and control the high-degree parity observable without replacing it by the independent surrogate.

6. Automatic sequences and exact recursions

The Rudin–Shapiro construction gives deterministic signs with square-root Fourier growth through an exact two-channel recursion [25]. This is one of the cleanest distant examples of deterministic square-root cancellation. It does not transfer by multiplication.

Proposition 6.1 (Untwisting obstruction)**.** *Let $r(n) \in \{-1, 1\}$ be any fixed sequence and define $b(n) = \mu(n)r(n)$. The identity*

$$\mathcal{M}(N) = \sum_{n\le N} b(n)r(n) \tag{6.1}$$

is exact. More generally, within the class of sign sequences, bounds $\sum_{n\le N} b(n) = O(\sqrt{N})$ and $\sum_{n\le N} r(n) = O(\sqrt{N})$ do not imply a square-root bound for $\sum_{n\le N} b(n)r(n)$.

Proof. The identity follows from $r(n)^2 = 1$. For the general non-implication, take $b(n) = r(n)$, with r equal to a Rudin–Shapiro sign sequence. Both unweighted sums are then $O(\sqrt{N})$, whereas $\sum_{n\le N} b(n)r(n) = N$. □

Müllner proved qualitative Möbius orthogonality for all automatic sequences [19]. This important theorem estimates $\sum \mu(n)r(n) = o(N)$; it does not provide the uniform square-root estimate and stable family reconstruction in Theorem 2.1. The exact Rudin–Shapiro recursion is therefore a model of the desired phenomenon, not a known engine for the constant mode.

7. Frames, restricted inverses, and discrepancy

Restricted invertibility, Kashin representations, and Kadison–Singer paving give strong stability and partition theorems [16, 20, 18]. Spencer's discrepancy theorem constructs a signing with discrepancy of square-root order [27]. These results solve different quantifier problems.

A fixed frame satisfying frame bounds gives stable ℓ^2 reconstruction for vectors in its span. Discrepancy theory states that, for suitable set systems, there exists a favorable coloring. The Möbius problem instead presents one prescribed coloring and asks for its correlation with one fixed vector. Neither existence theorem makes that coloring selectable.

Proposition 3.6 identifies the norm that matters. In a unit-norm orthonormal basis every representation of $\mathbf{1}_N$ has coefficient ℓ^2-norm $\sqrt{N}$ and hence ℓ^1-norm at least $\sqrt{N}$. For the Fourier basis the constant mode is a single coefficient, so a uniform bound that includes it already contains the desired estimate. Redundancy may improve robustness, but no theorem in the audited collection simultaneously supplies

$$\mathbf{1}_N = \sum_j c_j u_j, \qquad \|c\|_1 = N^{o(1)}, \qquad \sup_j |\langle \mu, u_j\rangle| \ll N^{1/2+\varepsilon} \tag{7.1}$$

under a normalization where the last two bounds combine to (1.2).

Stable signed sampling remains the closest audited framework to the logical shape of Theorem 2.1. It is not an achieved transfer because the deterministic Möbius correlation estimate in (7.1) is absent.

The lower bound in Proposition 3.6 does not by itself establish circularity. It constrains scale, not direction.

Proposition 7.1 (Large kernel norm does not imply a constant component)**.** *For every $N \ge 2$, there is a vector $u \in \mathbb{C}^N$ such that*

$$|u_n| = 1, \qquad \|u\|_2 = \sqrt{N}, \qquad \langle u, \mathbf{1}_N\rangle = 0. \tag{7.2}$$

Thus the condition $L \ge N^{1/2-o(1)}$, forced when $\|c\|_1 = N^{o(1)}$, does not force an individual kernel to be close to the constant vector.

Proof. Take $u_n = \exp(2\pi i n/N)$ for $1 \le n \le N$. The norm assertion is immediate, and the inner product vanishes by the finite geometric-series identity. □

Exact reconstruction does impose a separate directional constraint on the family as a whole.

Proposition 7.2 (Necessary constant projection)**.** *If $\mathbf{1}_N = \sum_j c_j u_j$, then*

$$\max_j |\langle u_j, \mathbf{1}_N\rangle| \ge \frac{N}{\|c\|_1}. \tag{7.3}$$

If additionally $\|u_j\|_2 \le L$, at least one kernel satisfies

$$\frac{|\langle u_j, \mathbf{1}_N\rangle|}{\|u_j\|_2 \|\mathbf{1}_N\|_2} \ge \frac{\sqrt{N}}{L\|c\|_1}. \tag{7.4}$$

Thus $L \asymp \sqrt{N}$ and $\|c\|_1 = N^{o(1)}$ forbid every kernel from having polynomially small constant projection, but they do not force a kernel to equal or converge in angle to the constant vector.

Proof. Pair the reconstruction with $\mathbf{1}_N$. Then

$$N = \left| \sum_j c_j \langle u_j, \mathbf{1}_N \rangle \right| \le \|c\|_1 \max_j |\langle u_j, \mathbf{1}_N \rangle|, \tag{7.5}$$

which proves (7.3). Divide by $L\sqrt{N}$ to obtain (7.4). □

There is also no gain from rescaling alone. Replacing u_j by $\alpha_j u_j$ and c_j by c_j/α_j leaves each product

$$|c_j|\,|\langle \mu, u_j \rangle| \tag{7.6}$$

unchanged. The possible leverage must therefore come from a structured directional class of oscillatory kernels, not from their size.

8. Transfer operators and deterministic dynamics

Under the regularity, spectral, and characteristic-function hypotheses of the relevant theorem, transfer-operator methods yield central-limit theorems and almost-sure invariance principles for centered observables. Gouëzel's spectral method is representative and supplies precise Brownian approximations under such hypotheses [11].

Every bounded sequence can be encoded formally as an orbit in a shift space: take $x = (a_1, a_2, \ldots)$, let T be the shift, and let $f(x) = x_1$. This formal encoding supplies no spectral gap and no typicality of the chosen orbit.

Proposition 8.1 (Centering in an orbit model)**.** *Suppose a probability-preserving system (X, T, π), an observable $f \in L^1(\pi)$, and a point x satisfy*

$$f(T^{n-1}x) = \mu(n), \qquad n \ge 1. \tag{8.1}$$

If x is generic for π with respect to f, then

$$\int f \, d\pi = \lim_{N\to\infty} \frac{\mathcal{M}(N)}{N}, \tag{8.2}$$

whenever the limit exists. A central-limit or invariance principle applying only to π-almost every point does not imply a bound for this specified point x.

Proof. Genericity is the equality between the Birkhoff average along x and the space average. Using (8.1) gives (8.2). An almost-everywhere statement may exclude any prescribed point of a nonatomic system, so it cannot by itself be specialized to x. □

Genericity of the specified Möbius orbit is not retained as a research route: proving it with the required rate merely moves the desired discrepancy estimate into the hypothesis. A logically different dynamical route would have to be uniformly deterministic. It would begin with a state space and local transition rules defined independently of the values of $\mathcal{M}(N)$, encode the Möbius path by those rules, and prove the required bound for every admissible initial state and every path in a broader locally specified class. A uniform cocycle contraction or a bounded coboundary obtained from local operator identities would meet this logical requirement. An almost-everywhere central limit theorem would not. No audited transfer-operator theorem currently provides this stronger uniform statement.

9. Empirical processes and dependent sampling

KMT strong approximation couples centered iid partial sums and empirical processes to Gaussian processes with exceptionally small error [14]. Modern dependent versions assume mixing, coupling, or functional-dependence conditions. Proposition 3.2 gives the decisive audit: sampling a fixed finite population controls

$$S_m - \frac{m}{N}\mathcal{M}(N), \tag{9.1}$$

not $\mathcal{M}(N)$ itself. At $m = N$ the centered expression is exactly zero. Increasing the sample size to all 455 million stored observations improves estimation of a finite empirical mean; it does not convert a confidence statement into a theorem about all future Möbius values.

The same issue persists for Markov and block-bootstrap prototypes. They are appropriate for inference under a declared stochastic sampling model, but the model assumption is the missing bridge in a pointwise deterministic proof.

The same obstruction can be written as an exact identity for cyclic windows. This is useful because it prevents an uncentered second-moment estimate from being mistaken for an independent bridge theorem.

Proposition 9.1 (Exact window-energy decomposition)**.** *Let $a_1, \dots, a_N \in \mathbb{C}$, extend the sequence periodically to $\mathbb{Z}/N\mathbb{Z}$, and let $1 \le H \le N$. For $x \in \mathbb{Z}/N\mathbb{Z}$, set*

$$S_{N,H}(x) = \sum_{h=0}^{H-1} a_{x+h}, \qquad A_N = \sum_{n=1}^{N} a_n. \tag{9.2}$$

Then

$$\sum_{x\in\mathbb{Z}/N\mathbb{Z}} |S_{N,H}(x)|^2 = \frac{H^2}{N}|A_N|^2 + \sum_{x\in\mathbb{Z}/N\mathbb{Z}} \left| S_{N,H}(x) - \frac{H}{N}A_N \right|^2. \tag{9.3}$$

In particular, for $a_n = \mu(n)$, every upper bound for the uncentered energy in the left-hand side also bounds the nonnegative term $H^2|\mathcal{M}(N)|^2/N$; it does not estimate that term by an independent fluctuation mechanism.

Proof. Each a_n occurs in exactly H cyclic windows, and hence

$$\frac{1}{N}\sum_{x\in\mathbb{Z}/N\mathbb{Z}} S_{N,H}(x) = \frac{H}{N}A_N. \tag{9.4}$$

The constant vector and the mean-zero vector $\left(S_{N,H}(x) - HA_N/N\right)_x$ are orthogonal in $\ell^2(\mathbb{Z}/N\mathbb{Z})$. The Pythagorean identity gives (9.3). □

For example, Cauchy–Schwarz applied to (9.4) gives only

$$|A_N|^2 \le \frac{N}{H^2}\sum_{x\in\mathbb{Z}/N\mathbb{Z}} |S_{N,H}(x)|^2, \tag{9.5}$$

which merely extracts the constant-mode term already displayed in (9.3). At $H = N$, every cyclic window is identically A_N, and the left-hand side equals $N|A_N|^2$. Thus the energy bound $\sum_x |S_{N,N}(x)|^2 \ll N^2$ is equivalent to $|A_N| \ll N^{1/2}$. For $H < N$, periodic wraparound is an artificial boundary convention. With genuine intervals, the corresponding first-moment identity has nonuniform boundary weights, so (9.4) cannot be invoked unchanged. Any theorem formulated only after subtracting HA_N/N concerns the second term on the right of (9.3); by construction it says nothing about the first term.

10. Random multiplicative chaos and decoupling

Random multiplicative functions display subtle cancellation and multiplicative-chaos behavior. Harper determined low moments at scales even smaller than the naive square-root prediction in the first absolute moment [12], and Soundararajan–Xu proved central-limit theorems for selected sums of Steinhaus random multiplicative functions [26]. Decoupling inequalities systematically replace suitable dependent random chaoses by independent counterparts [5].

These results average over the prime phases. The deterministic Möbius choice corresponds to one prescribed configuration of those phases, together with the squarefree gate. Proposition 3.3 shows that an ambient L^p moment bound alone does not control evaluation at that configuration. In a finite Rademacher model the unrestricted evaluation norm is exponential in the number of coordinates, as recorded in Corollary 3.5. This does not exclude a sharper inequality on a restricted arithmetic function class, but no such uniform-in-phase theorem is supplied by the cited moment or decoupling results. The audited transfer therefore lacks a deterministic-evaluation bridge.

11. Function fields

Function fields provide exact square-root phenomena and, in many problems, complete analogues of conjectural integer behavior. For the full set of monic polynomials, the polynomial Möbius sum is even simpler.

Theorem 11.1 (Exact full-degree cancellation over a polynomial ring)**.** *Let μ_q be the polynomial Möbius function. Then*

$$\sum_{\substack{F\in\mathbb{F}_q[t]\ \text{monic}\\ \deg F=n}} \mu_q(F) = \begin{cases} 1, & n=0,\\ -q, & n=1,\\ 0, & n\geq 2. \end{cases} \tag{11.1}$$

Proof. Unique factorization gives the formal Euler product

$$\sum_{F\ \text{monic}} \mu_q(F)u^{\deg F} = \prod_{P\ \text{monic irreducible}} (1-u^{\deg P}). \tag{11.2}$$

The zeta series of $\mathbb{F}_q[t]$ is $\sum_{F\ \text{monic}} u^{\deg F} = (1-qu)^{-1}$. Taking its reciprocal in (11.2) gives $1-qu$, and coefficient comparison proves (11.1). □

Keating and Rudnick computed means and variances for short intervals and arithmetic progressions in $\mathbb{F}_q[t]$, using finite-field equidistribution and unitary matrix integrals [13]. Those theorems are genuine and often reach square-root scales. No known specialization sends polynomial degree boxes to integer initial intervals while preserving Möbius values and uniform constants. The tempting formal limit $q\to 1$ is not such a map. The function-field route therefore offers a solved comparison model, not an existing transfer theorem to (1.1).

12. Numerical audit of the Boolean model

Experiment 12.1 (Reported Boolean-surrogate computation)**.** The computational record states that the binary file contains 455,052,511 consecutive int8 Möbius values beginning at $\mu(1)$. It also states that the accompanying linear Python script memory-maps the file, computes exact prefix sums, and uses a segmented sieve to evaluate

$$B(N) = N\prod_{p\leq N}(1-p^{-1})^2 \tag{12.1}$$

at logarithmically spaced prefixes. Set

$$N_* = 455{,}052{,}511. \tag{12.2}$$

At the final prefix the reported values are

$$\mathcal{M}(N_*) = -5414, \qquad \frac{|\mathcal{M}(N_*)|}{\sqrt{N_*}} = 0.25379759, \tag{12.3}$$

whereas the independent surrogate gives

$$B(N_*) = 360{,}930.55398, \qquad \frac{B(N_*)}{\sqrt{N_*}} = 16.91970904. \tag{12.4}$$

Figure 1 displays the reported normalized comparison. The computation gives no asymptotic estimate for $\mathcal{M}(N)$ and is not a statistical rejection test. It records a substantial finite-scale mismatch between the observed prefix and the mean of the unrestricted independent surrogate.

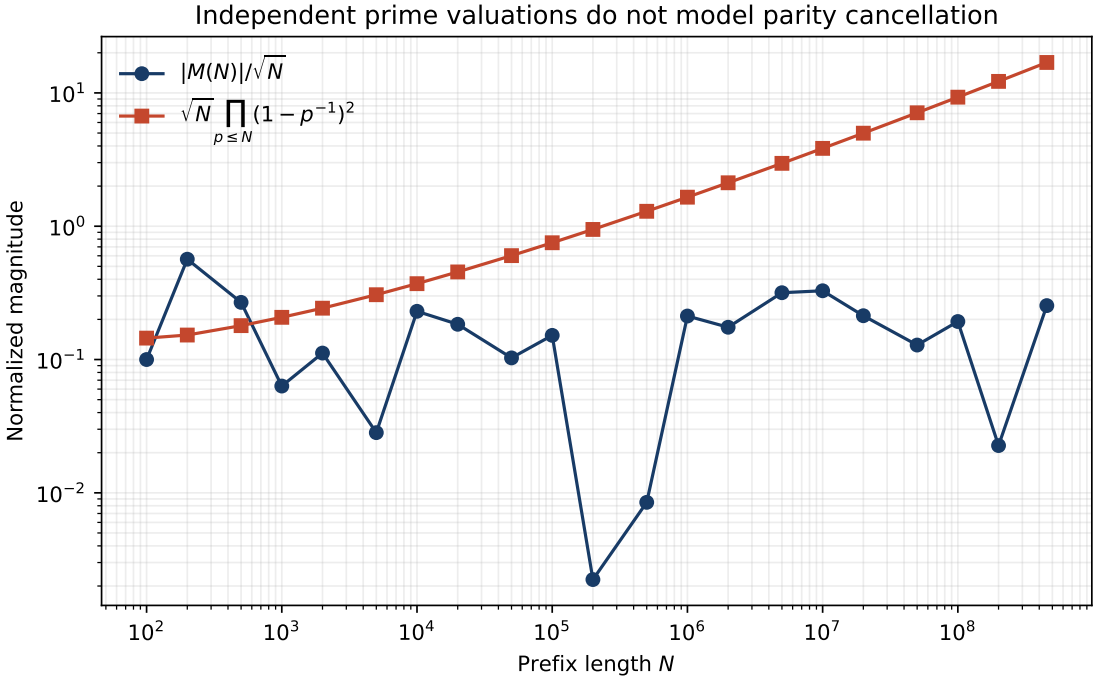


FIGURE 1. Reported stored Möbius prefixes versus the mean predicted by the unrestricted independent p-adic surrogate. The comparison records finite-scale model mismatch and is not used as evidence for an asymptotic pointwise theorem.

13. AUDIT VERDICTS

TABLE 1. Uniform audit of distant square-root theories.

Area	Genuine theorem	Missing interface	Audit conclusion
Martingales and Stein	Square-root fluctuations and concentration for centered random processes	Conditional centering or regression is unavailable for the fixed Möbius path	No applicable theorem identified
Boolean hypercontractivity	Sharp product-space norm inequalities	High-degree parity and the unrestricted independent surrogate has mean $N/(\log N)^2$	Surrogate mismatch; no transfer
Automatic sequences	Exact Rudin–Shapiro square-root recursion	Untwisting requires a second weighted estimate; known Möbius orthogonality is qualitative	No square-root untwisting theorem identified
Frames and stable sampling	Restricted inverses, Kashin representations, paving	The direct argument needs ℓ^1 control and a uniform deterministic correlation estimate	No complete stable-transfer theorem identified
Discrepancy	Existence of a square-root-discrepancy coloring	The theorem chooses the signs; Möbius signs are prescribed	Wrong quantifier for this application
Transfer operators	CLT and ASIP under spectral hypotheses	Spectral gap, centering, and deterministic control of the fixed arithmetic orbit are unproved	No uniform fixed-orbit theorem identified
Empirical processes	KMT-scale approximation of centered samples	Controls deviation from $\mathcal{M}(N)/N$, not that mean	Centers away the target
Random multiplicative chaos	Sharp moment and distribution theorems	Ambient L^p control does not bound a prescribed phase	Deterministic-evaluation bridge absent
Decoupling	Comparison of dependent and independent random chaos	Requires a probability-space coupling and does not control a prescribed phase	No deterministic specialization identified
Function fields	Exact cancellation and square-root variance theorems	No specialization preserving intervals, Möbius values, and constants	Solved analogue; no transfer map

No entry in Table 1 is admissible under Theorem 2.1. This statement concerns the audited theorem set; it is not a claim that every conceivable theorem in these fields is incapable of contributing.

14. Closure of the conditional templates and the involution reformulation

14.1. **Why the two conditional templates are not retained.** The independent-transfer conditions used in the previous version were necessary but not sufficient. Requiring a theorem to hold on a class larger than the single Möbius instance does not create an independent proof of that theorem. In the frame program, the decisive input remains the uniform bound

$$\sup_j |\langle \mu, u_j \rangle| \ll_\varepsilon N^{1/2+\varepsilon} \tag{14.1}$$

for a reconstructing family with substantial low-frequency content. The frame theorems control reconstruction norms, not the arithmetic correlations in (14.1). Proposition 7.2 also shows that stable reconstruction prevents every normalized kernel from being uniformly remote from the constant direction. After the known analytic correlation machinery has been excluded by the scope of this audit, no independent engine remains for (14.1). The frame program is therefore closed within this scope, rather than retained as a conditional route.

The same correction applies to deterministic transfer operators. A spectral gap for an abstract operator gives no estimate for a prescribed arithmetic path. Requiring contraction for every path in a class containing the Möbius path would be stronger, but the proof of that contraction is precisely the missing arithmetic input. If it is verified through correlations of the Möbius path, the argument is circular; if it is not, no applicable theorem is currently available. Calling the class "uniform" or "locally defined" does not resolve this interface. Thus the transfer-operator program is also closed within the declared search.

This conclusion is scoped. It does not prove that frame theory or dynamical systems can never participate in a future proof. It says that they supply no presently independent square-root mechanism once the classical arithmetic engines capable of proving uniform twisted estimates are removed.

14.2. **A sign-reversing involution program.** Set

$$\mathcal{S}_N = \{n \le N : \mu(n)^2 = 1\}, \qquad \mathcal{S}_N^{\pm} = \{n \in \mathcal{S}_N : \mu(n) = \pm 1\}. \tag{14.2}$$

The following elementary identity describes a different proof format: deterministic cancellation in explicit pairs, rather than a uniform correlation estimate. It does not by itself make the problem easier.

Proposition 14.1 (Exact cancellation by an involution)**.** *Let $E_N \subseteq \mathcal{S}_N$, and let $T_N : \mathcal{S}_N \setminus E_N \to \mathcal{S}_N \setminus E_N$ satisfy*

$$T_N(T_N(n)) = n, \qquad T_N(n) \neq n, \qquad \mu(T_N(n)) = -\mu(n). \tag{14.3}$$

Then

$$\mathcal{M}(N) = \sum_{n \in E_N} \mu(n), \qquad |\mathcal{M}(N)| \le |E_N|. \tag{14.4}$$

Proof. Terms with $\mu(n) = 0$ do not contribute to $\mathcal{M}(N)$. The involution partitions $\mathcal{S}_N \setminus E_N$ into two-element orbits. The two Möbius values in each orbit sum to zero by (14.3). Only the terms in E_N remain, which gives both assertions in (14.4). □

14.3. **What successful involutions do in other settings.** There are rigorous precedents for the architecture in Proposition 14.1. The closest elementary arithmetic example is

$$\sum_{d\mid m}\mu(d)=0 \qquad (m>1). \tag{14.5}$$

After zero terms are removed, fix one prime $p \mid m$ and toggle whether p divides d. The squarefree divisors of m form a complete Boolean cube, the toggle never leaves that cube, and every pair has opposite Möbius sign. The interval condition $d \le N$ destroys precisely this closure property when one tries to apply the same operation to $\mathcal{S}_N$.

Franklin's involution for Euler's pentagonal-number theorem is a more informative model; see Pak [24]. If $d_e(t)$ and $d_o(t)$ count partitions of t into distinct parts with respectively even and odd numbers of parts, the involution gives

$$d_e(t)-d_o(t)=\begin{cases}(-1)^r, & t=r(3r-1)/2 \text{ for some } r\in\mathbb{Z},\\ 0, & \text{otherwise.}\end{cases} \tag{14.6}$$

Its mechanism has four components: a conserved grade t, a canonical local move, reversibility of that move, and a complete classification of the fixed points. Only generalized pentagonal shapes are fixed, so the number of unpaired objects with grade at most X is $O(\sqrt{X})$. The general involution principle systematizes this style of cancellation [9]. Equation (14.6) is therefore a genuine example of square-root-sparse exceptions, but its conserved additive weight and rigid normal forms have no known counterpart for products of distinct primes.

Discrete Morse theory gives a geometric version of the same philosophy: an acyclic matching on the face poset pairs incident cells of consecutive dimensions, and unmatched critical cells control the resulting cell complex [7]. The analogy is exact for Forman acyclic matchings; it is not an identification of every sign-reversing involution with a discrete Morse matching. Within the Forman class it supplies a decisive obstruction. Define

$$\Delta_N=\{P(n): n\le N \text{ is squarefree}\}, \qquad P(n)=\{p : p\mid n\}. \tag{14.7}$$

This is a simplicial complex because divisors of a squarefree integer not exceeding N also do not exceed N. Björner proved

$$\mathcal{M}(N)=-\widetilde{\chi}(\Delta_N)=\sum_{k\ge 0}(-1)^{k-1}\beta_k(\Delta_N), \tag{14.8}$$

and, for every $\theta > 17/54$,

$$\begin{aligned}\sum_{k\ge 0}\beta_k(\Delta_N)&=\frac{2N}{\pi^2}+O_\theta(N^\theta),\\ \sum_{k\equiv 0 \pmod 2}\beta_k(\Delta_N)&\sim\sum_{k\equiv 1 \pmod 2}\beta_k(\Delta_N)\sim\frac{N}{\pi^2}.\end{aligned} \tag{14.9}$$

See [3]. By the weak Morse inequalities, every Forman discrete Morse function on Δ_N has at least $\beta_k(\Delta_N)$ critical k-cells. Hence its total number of critical cells is bounded below by a positive constant times N, not by $N^{1/2+\varepsilon}$. Standard discrete Morse matching therefore cannot supply the desired sparse exceptional set. The cancellation in (14.8) occurs between two quantities of linear size and is exactly the unresolved arithmetic cancellation. Thus the Euler-characteristic formulation is an established translation, not a new route to the target.

An odd-block involution would not be a Forman matching: it may replace several vertices at once and need not pair incident faces. This departure is necessary to escape (14.9), but it also forfeits the Morse-theoretic machinery that proves cancellation and controls critical cells in other complexes.

The simplest multiplicative graph cannot achieve the target. Let $G_N^{(1)}$ have vertex set $\mathcal{S}_N$, with an edge between n and pn whenever p is prime, $p \nmid n$, and $pn \le N$. Every edge reverses the Möbius sign.

Proposition 14.2 (Large-prime obstruction to one-prime pairing)**.** *Every matching in $G_N^{(1)}$ leaves at least*

$$\pi(N) - \pi(N/2) - 1 \tag{14.10}$$

vertices unmatched. Consequently, one-prime involutions have an exceptional set bounded below by a positive constant multiple of $N/\log N$ for all sufficiently large N.

Proof. If ℓ is prime and $N/2 < \ell \le N$, then ℓ cannot be multiplied by any prime while remaining at most N. If $\ell = pn$ with p prime, then primality forces $n = 1$. Thus every such ℓ has the unique neighbor 1 in $G_N^{(1)}$. A matching can pair at most one of these primes with 1, proving (14.10). The prime number theorem gives the final order statement [17]. □

This obstruction is consistent with the appearance of the scale $N/\log N$ in the ordinary divisor graph; see Melotti and Saias [22]. It forces any surviving involution to use more than single-prime divisibility edges.

Definition 14.3 (Odd-block multiplicative move)**.** For $n, m \in \mathcal{S}_N$, write $P(n)$ for the set of prime divisors of n. An odd-block move from n to m is a replacement

$$\begin{aligned} m &= n\frac{\prod_{p\in A} p}{\prod_{q\in B} q}, \quad A \cap P(n) = \varnothing, \\ B &\subseteq P(n), \qquad\quad |A| + |B| \equiv 1 \pmod 2. \end{aligned} \tag{14.11}$$

It is multiplicatively C-balanced when

$$C^{-1} \le \frac{\prod_{p\in A} p}{\prod_{q\in B} q} \le C. \tag{14.12}$$

The parity condition in (14.11) gives $\mu(m) = -\mu(n)$. In particular, a large prime may be paired with a product of two smaller primes by replacing one prime with two, a three-coordinate change that is unavailable in $G_N^{(1)}$.

The unproved search target is an explicit involution made from balanced odd-block moves for which, for every $\varepsilon > 0$,

$$|E_N| \ll_\varepsilon N^{1/2+\varepsilon}. \tag{14.13}$$

Proposition 14.1 would then give (1.2) immediately. Equation (14.13) is not proved here.

The construction is noncircular only if the pairing rule is specified from prime supports and fixed scale partitions before $\mathcal{M}(N)$ is evaluated, its injectivity is proved directly, and its exceptional set is counted through unsigned factorization conditions. Merely computing a maximum matching, or invoking Hall's theorem [21] without proving its neighborhood inequalities independently of the sign imbalance, relocates the original problem. Indeed, (14.4) implies the unavoidable lower bound $|E_N| \ge |\mathcal{M}(N)|$ for every sign-reversing partial involution. The only possible new content would therefore be an explicit multiscale pairing whose exceptional set can be bounded using only the geometry of prime factorizations. No such square-root construction, partial exponent improving the one-prime barrier, or theorem suggesting that it exists is presently supplied. More importantly, any estimate $|E_N| \le B(N)$ for a sign-reversing involution immediately gives $|\mathcal{M}(N)| \le B(N)$ by Proposition 14.1. The exceptional-set theorem is therefore not an intermediate transfer theorem: it is a direct sufficient condition for the corresponding Mertens bound.

An earlier version proposed $o(N/\log N)$ as a first exceptional-set target. That target is withdrawn because it is weaker than the known unconditional estimate

$$\mathcal{M}(N) \ll N \exp\Big\{-c(\log N)^{3/5}(\log\log N)^{-1/5}\Big\} \qquad (N \geq N_0), \tag{14.14}$$

for suitable constants $c > 0$ and N_0; explicit versions are given by Lee and Leong [15]. A fixed-power exceptional-set estimate $|E_N| \ll N^{1-\delta}$ would instead constitute a major new pointwise bound for $\mathcal{M}(N)$ through the same elementary identity. It cannot be advertised as a modest preliminary objective.

The parity of the number of prime factors is precisely the information hidden by the classical sieve parity phenomenon. Friedlander and Iwaniec's asymptotic sieve overcomes that phenomenon in special sequences by adding bilinear information [8]. If an exceptional-set estimate requires such parity-sensitive cancellation, the proposed combinatorial language has returned to the analytic machinery excluded from this audit. The odd-block proposal is therefore closed as a forward program, while its exact formulation remains in the report as a record of the obstruction.

Automatic paraunitary recursions are removed from the forward program. Their exact square-root mechanism controls the recursively generated automatic sequence itself, while Proposition 6.1 shows that inserting and then removing an automatic weight requires a second uncontrolled correlation. Müllner's qualitative orthogonality theorem supplies no missing stable inverse or second-channel estimate. This does not constitute an impossibility theorem, but it leaves no noncircular transfer mechanism to pursue.

Function-field comparison is also removed. The exact analogues remain useful as controls, but no candidate specialization currently preserves the interval, the Möbius values, and subpolynomial constants. Without a proposed normed map, the analogy does not define a research route that can be audited by Theorem 2.1.

15. Conclusion

The audit found many exact square-root theorems far from classical analytic number theory, but no existing theorem that can presently be transferred to $\mathcal{M}(N)$ at square-root scale. The failure mechanisms are now explicit: unknown centering for martingales and empirical processes, high-degree parity misrepresented by product probability, untwisting for automatic sequences, ℓ^1 loss in the direct frame argument, choice of signs in discrepancy, exceptional fixed orbits for transfer operators, absence of an ambient-to-point estimate for random chaos, and absence of a normed specialization from function fields.

The exact-transfer lemma remains a strict diagnostic for future transference candidates. It does not, however, manufacture the decisive uniform correlation estimate. The earlier decision to retain signed frames and uniform deterministic operators mistook a well-posed missing hypothesis for an independent mechanism. Both are now closed within the deliberately distant scope of this audit. Automatic recursions and function-field analogy likewise remain only as negative controls.

No forward route in the audited collection survives under the declared exclusions. The sign-reversing-involution section does not alter that verdict. Proposition 14.2 proves that its naive local version leaves order $N/\log N$ vertices unmatched, while Proposition 14.1 shows that any exceptional-set estimate immediately implies the corresponding pointwise Mertens bound. The previously proposed $o(N/\log N)$ milestone has been removed because it is weaker than the unconditional estimate (14.14).

Nor does cyclic-window energy provide a bridge. The exact decomposition in Proposition 9.1 shows that the uncentered energy already contains the desired constant mode as a nonnegative orthogonal component; centering removes that component rather than estimating it. Thus the window argument is another exact reformulation, not a source of cancellation. Within the audited collection, no completed transfer route or independently proved intermediate estimate remains. This scoped conclusion is not an impossibility theorem for future mathematics outside that collection.

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